
Dynamical-phase amplification limits the noise resilience of adaptive variational Berry-phase estimation

Anonymous

Abstract

Adaptive variational quantum dynamics compresses the cyclic adiabatic evolution needed for a many-body Berry phase into circuits orders of magnitude shallower than a Trotterised loop, but the accuracy of the resulting topological invariant on a noisy device has not been established. We characterise it. The observable is a sum of two separately measured pieces, an interferometric phase φ_{qc} and a classically integrated global phase $\varphi_G = \varphi_{G,1} + \varphi_{G,2}$ built from device-measured energies and connection overlaps, and we inject a scheduled, device-referenced noise model — two-qubit depolarising, single-qubit depolarising, asymmetric readout error and T_1/T_2 relaxation, plus genuine finite-shot sampling — into *every* circuit the method executes. Four things follow. First, the interferometric leg is far more robust than it looks: reading both quadratures of the ancilla makes φ_{qc} exactly invariant under any common contrast loss, so unital noise costs it variance but no bias. Second, the robustness does not transfer to $\varphi_{G,1}$: the dynamical phase cancels between the two halves of the time-reversed loop only after two half-loop integrals of 53 and 213 rad cancel to $O(1)$, and any depth-asymmetric distortion of the measured energy is amplified by that same factor, making the dynamical leg the dominant error channel. Third, the resulting thresholds are strongly depth-dependent: on a superconducting-class profile the non-interacting chain tolerates two-qubit error up to $\epsilon_{2\text{q}} \simeq 6.1 \times 10^{-4}$ while the interacting chain, whose circuits are $6.6\times$ deeper, breaks at $\epsilon_{2\text{q}} \simeq 7.6 \times 10^{-6}$. Fourth, finite sampling is a separate and irreducible axis whose floor is set by the same amplification. A noise-estimation-circuit rescaling aimed at the dynamical leg raises every threshold by $1.5\text{--}2.1\times$, and at a common noiseless accuracy target the shallow variational loop tolerates up to $30\times$ more noise than a Trotterised loop of the same task at $U = 0$ — but *less* at $U = 10$, in both topological regimes, because there the amplified dynamical leg, not the circuit depth, is what fails. Geometric robustness, in this algorithm, protects the geometric phase and not the bookkeeping that surrounds it.

1 Introduction

The Berry phase of a many-body ground state transported around a closed loop in parameter space is a gauge-invariant, quantised diagnostic of topological order [1, 2, 3, 4]. For a one-dimensional, time-reversal-invariant chain threaded by a twist it takes the value 0 in a trivial phase and π in a topologically non-trivial one, and for interacting electrons it is exactly the quantity that resists efficient classical computation [5, 6]. Quantum algorithms that obtain it by adiabatically transporting the ground state around the loop and reading the accumulated phase off an ancilla [7, 8] are therefore an attractive near-term target, and adaptive variational quantum dynamics simulation (AVQDS) [9, 10] makes them dramatically cheaper: rather than Trotterising the propagator, it grows a problem-specific ansatz whose McLachlan distance to the exact evolution stays below a threshold, compressing a $10^5\text{--}10^6$ CNOT loop into a few hundred gates [11].

That compression is the entire practical argument for the method, and it is an argument about noise. It has not, however, been tested against noise. The adaptive variational Berry-phase pipeline has been characterised only under exact arithmetic, where it is remarkably forgiving: the loop returns the correct quantised phase even when the mid-loop fidelity with the instantaneous ground state has fallen to 0.5, because the phase is a closed-loop, gauge-invariant quantity and the fast dynamical phases of the diabatic admixtures average out [11]. Robustness to non-adiabaticity, though, is a statement about a coherent, deterministic error, and it says nothing about stochastic device noise; the two are physically different, and the literature on geometric phases contains both strong claims of intrinsic protection [12, 13, 14, 15] and careful demonstrations that geometric and dynamical gates can be equally noise-sensitive [16].

There is a structural reason to expect the answer to be subtle here. Because the variational propagation is generated in a density-matrix (projective) formulation, it does not track the global phase, and the reported Berry phase is the sum of two separately obtained quantities,

$$\varphi_B = \varphi_{qc} + \varphi_G, \quad \varphi_G = \varphi_{G,1} + \varphi_{G,2}, \quad \varphi_{G,1} = -\int \langle \hat{\mathcal{H}} \rangle_\theta dt, \quad \varphi_{G,2} = \int \sum_\mu a_\mu d\theta_\mu, \quad (1)$$

where φ_{qc} comes from an ancilla interferometer and φ_G from a classical quadrature of quantities the device must measure — the energy expectation and the geometric connection overlaps a_μ . These three legs are measured on different circuits. Their deterministic biases may partially cancel, because they share one frozen trajectory; their stochastic fluctuations cannot, because they are drawn independently. And $\varphi_{G,1}$ is not like the other two: it is a *dynamical* phase whose two halves are individually enormous (53 and 213 rad for the two interaction strengths studied here) and cancel to $O(1)$ only by time-reversal symmetry. Any perturbation of the measured energy that is not symmetric between the two halves of the loop is amplified by that ratio before it reaches the Berry phase.

Contributions. We turn this into a quantitative characterisation.

- We re-implement the adaptive variational Berry-phase pipeline from scratch and reproduce its noiseless behaviour — quantised phases across the whole dimerisation axis, the loop-resolved infidelity trace, and the circuit-resource profile — against the authors’ published traces (Sec. 4).
- We define a scheduled, device-referenced noise model and inject it into *every* circuit the method runs: the two quadratures of the ancilla interferometer, the energy-measurement circuits, and one connection-overlap circuit per variational parameter at every quadrature node (Sec. 3). Finite-shot records are drawn from the resulting exact output distributions.
- We show that the natural two-quadrature estimator makes φ_{qc} exactly invariant under a common contrast loss, and we extract noise thresholds in device-referenced units for four configurations, with bootstrap uncertainties and a separately reported shot floor (Sec. 5).
- We decompose the error into its three legs and isolate the four noise channels. In the interacting regime — the one the method exists for — the dynamical leg $\varphi_{G,1}$ alone accounts for the failure; in the shallow non-interacting regime the two global-phase legs partially cancel and the assembled phase outlives both of them (Sec. 6). We quantify the depth dependence between the two, and find that the dominant channel changes with it: readout for the shallow circuits, T_1/T_2 relaxation for the deep ones.
- We show that a noise-estimation-circuit rescaling aimed at the dominant leg raises every threshold, that the shallow variational loop tolerates up to $30\times$ more noise than a Trotterised loop at $U = 0$ but *less* at $U = 10$ in both topological regimes — because there the bookkeeping, not the depth, is the binding constraint — and how the picture changes on a trapped-ion-class profile (Sec. 8).

2 The algorithm and the test bed

Model. We study the Su–Schrieffer–Heeger–Hubbard chain [17, 18] with $N = 4$ sites and periodic boundary conditions,

$$\hat{\mathcal{H}} = \sum_{j\sigma} \left[t_{j,j+1} \hat{c}_{j+1\sigma}^\dagger \hat{c}_{j\sigma} + \text{h.c.} \right] + U \sum_j \hat{n}_{j\uparrow} \hat{n}_{j\downarrow} - \frac{U}{2} \sum_{j\sigma} \hat{n}_{j\sigma}, \quad t_{j,j+1} = t(1 + (-1)^j \delta), \quad (2)$$

with $t = 1$ and the twist imposed on the spin-up species only, $\hat{c}_{j+N\uparrow} = e^{i\lambda}\hat{c}_{j\uparrow}$, so that the Berry phase tracks the topological transition at $\delta = 0$ rather than being locked to multiples of 2π . A Jordan–Wigner transformation in the interleaved mode order $m(j, \sigma) = 2j + \sigma$ maps the model onto $N_q = 8$ qubits and produces $N_H = 18$ Pauli strings for $U = 0$ (twelve of weight three, six of weight seven) and $N_H = 22$ for $U = 10$; this census is a first check that our independent construction matches the original.

Pipeline. We hold the algorithm fixed. The ground state at $\lambda = 0$ is prepared by adaptive variational imaginary-time evolution (AVQITE) [19] from the UCCSD-like pool of $N_p = 616$ generators (products of two or four $\hat{\sigma}^{x,y}$ with an odd number of $\hat{\sigma}^y$). The cyclic evolution is carried out by AVQDS with the Hamiltonian Pauli strings as the operator pool. In both cases the ansatz $|\Psi(\theta)\rangle = \prod_{\mu} e^{-i\theta_{\mu}\hat{A}_{\mu}}|\Psi_0\rangle$ is grown whenever the McLachlan distance exceeds $L_{\text{cut}}^2 = 10^{-4}$, appending a layer of disjoint-support generators in order of increasing L^2 [20], and the parameters follow $\sum_{\nu} M_{\mu\nu}\dot{\theta}_{\nu} = V_{\mu}$ integrated by RK4 with an adaptive step obeying $\max_{\mu} |\dot{\theta}_{\mu} \delta t| < \delta\theta_{\text{max}} = 0.01$. The loop is the time-reversed one: $t : 0 \rightarrow T/2$ with $\lambda : 0 \rightarrow \pi$, then $t : T/2 \rightarrow 0$ with $\lambda : \pi \rightarrow 2\pi$, which cancels the dynamical phase and doubles the geometric one. We use $T = 20$ throughout.

Readout. The interferometer measures $\langle \Psi_0 | U_G[\theta^1(0)]^{\dagger} U_{\text{full}}[\theta(2\pi)] | \Psi_0 \rangle = |\cdot| e^{i\varphi_{\text{qc}}}$ with one ancilla-controlled single-qubit rotation per variational parameter, and the global phase omitted by the projective propagation is restored through Eq. (1). Writing the physical state as $e^{i\varphi_G(t)}|\Psi[\theta(t)]\rangle$ and projecting $i\partial_t|\phi\rangle = \hat{\mathcal{H}}|\phi\rangle$ onto $\langle\Psi|$ gives $\dot{\varphi}_G = -\langle\hat{\mathcal{H}}\rangle_{\theta} + \sum_{\mu} a_{\mu}\dot{\theta}_{\mu}$ with $a_{\mu} \equiv \langle\hat{A}_{\mu}\rangle_{\phi_{\mu-1}}$ real, so that $\langle\Psi|\partial_{\theta_{\mu}}\Psi\rangle = -i a_{\mu}$. The a_{μ} are ordinary Pauli expectation values on the circuit *prefix* that precedes generator μ ; this is what makes them device-measurable, and it is the form we use.

Held fixed and disclosed deviations. The algorithm, the pool, the integrator, L_{cut}^2 , $\delta\theta_{\text{max}}$, the loop, the test bed and the readout identity are exactly those of the method being characterised. Two implementation choices require disclosure. (i) The linear system $M\dot{\theta} = V$ is rank deficient; we solve it with a Tikhonov shift $(M + \varepsilon_{\text{reg}}\lambda_{\text{max}}(M)\mathbb{K})\dot{\theta} = V$ at $\varepsilon_{\text{reg}} = 10^{-6}$ and evaluate the McLachlan distance as the *actual* residual $L^2 = \dot{\theta}^{\text{T}}M\dot{\theta} - 2V^{\text{T}}\dot{\theta} + 2\text{var}_{\theta}[\hat{\mathcal{H}}]$ of the $\dot{\theta}$ that is used. The commonly quoted form $L^2 = 2\text{var} - V^{\text{T}}\dot{\theta}$ is algebraically equivalent only at the exact minimiser; with any regularised solve it can report $L^2 \approx 0$ for a $\dot{\theta}$ whose true residual is large, silently switching the adaptive growth off. A scan over ε_{reg} from 10^{-4} to 10^{-8} is logged in `logs/regularization_scan.json`: the Berry phase is accurate throughout, but the ansatz size is not monotone — at $\varepsilon_{\text{reg}} = 10^{-4}$ the inflated L^2 drives the interacting ansatz to 817 generators, against 173 at our 10^{-6} . (ii) The reference product state prescribed for the ansatz (“spin-up in the first $N/2$ orbitals, spin-down in the remaining $N/2$ ”) fixes the occupation pattern but not the origin of the unit cell, and with the origin at site 0 the state has an exactly vanishing overlap with the ground state for $\delta > 0$. We shift the origin by one site in that regime, which is the same δ -dependent choice the published data files contain. Neither choice touches the algorithm; both are recorded in `attempts_log.md`.

3 Noise model, estimator and accuracy criterion

Everything in this section was fixed before any noisy result was produced; the machine-readable version is `logs/protocol.json`, written at pre-registration time.

3.1 Scope

The ansatz and the variational trajectory $\theta(t), \dot{\theta}(t)$ are frozen at their noiseless values, so the McLachlan integration is not re-run under noise. Every quantity the device must *measure* is re-evaluated under noise: both quadratures of the interferometer, the energy expectation feeding $\varphi_{G,1}$, and the connection overlaps feeding $\varphi_{G,2}$. The threshold we report is therefore a whole-method number at a fixed variational trajectory; the complementary study in which sampling noise is fed back into the McLachlan solve, so that the trajectory itself becomes stochastic, is outside our scope and is discussed in Sec. 11.

3.2 Circuits and channels

Each parametrised unitary $e^{-i\theta P}$ on a weight- p Pauli string is compiled, assuming all-to-all connectivity (the connectivity the method’s own resource model assumes), into single-qubit basis changes, a CNOT ladder, one R_z , the reversed ladder and the inverse basis changes: $2(p-1)$ CNOTs. In the ancilla-controlled interferometer the central rotation becomes $R_z(\theta) CX R_z(-\theta) CX$, i.e. one controlled rotation and two extra CNOTs per parameter. Generators with disjoint support are placed in the same layer, and layers are expanded into elementary time slices; in the controlled circuit the shared ancilla serialises the controlled rotations.

The state is a full $2^n \times 2^n$ density matrix ($n = 8$, or 9 with the ancilla) and every channel is a completely positive map applied to it:

- single-qubit depolarising with probability p_1 after every single-qubit gate, and two-qubit depolarising with probability p_2 after every CNOT;
- thermal relaxation (amplitude damping towards $|0\rangle$ plus pure dephasing) with times T_1, T_2 applied to the qubits of each gate for that gate’s duration and to every qubit for its residual idle time at the end of each layer;
- an asymmetric classical readout channel $P(1|0) = e_{01}$, $P(0|1) = e_{10}$ on every measured qubit, after which a finite number S of shots is drawn from the resulting distribution by multinomial (or binomial) sampling.

Nothing about the phase is perturbed by hand: the noise enters only as these maps on ρ , and the phase is then read off the resulting probabilities. The mixedness that results is the check that this is what actually happens (`logs/purity.json`). At $\eta = 0$ every circuit stays pure to 3×10^{-13} in $\text{Tr } \rho^2$; at the base rate $\eta = 1$ the full-loop interferometer ends at $\text{Tr } \rho^2 = 0.15$ for the 210-CNOT non-interacting circuit and 0.0024 for the 1674-CNOT interacting one, the latter only 24% above the maximally mixed floor $1/2^9 = 0.00195$ — the deep circuit is very nearly fully decohered, and its interferometric contrast has collapsed from 0.883 to 0.00835.

The base profile **SC-A** is superconducting-transmon class with all-to-all connectivity assumed: $p_2 = 2.0 \times 10^{-3}$, $p_1 = 2.5 \times 10^{-4}$, $e_{01} = 7.0 \times 10^{-3}$, $e_{10} = 1.3 \times 10^{-2}$, $T_1 = 200 \mu\text{s}$, $T_2 = 130 \mu\text{s}$, $\tau_{1q} = 32 \text{ ns}$, $\tau_{2q} = 68 \text{ ns}$; the rates follow published Heron-class calibration medians [21, 22]. The readout channel is deliberately asymmetric, $e_{10} > e_{01}$, because transmon readout error is dominated by $|1\rangle \rightarrow |0\rangle$ decay during the measurement window; this matters, because a symmetric readout error would contract both interferometer quadratures equally and cancel out of Eq. (3) entirely. The channel composition follows the validated depolarising+relaxation+readout model of [23]. A secondary profile **TI-B** uses trapped-ion-class numbers ($p_2 = 1.5 \times 10^{-3}$, $p_1 = 3 \times 10^{-5}$, SPAM $\sim 1.5 \times 10^{-3}$, $T_1 = 10 \text{ s}$, $T_2 = 3 \text{ s}$, $\tau_{2q} = 300 \mu\text{s}$), taken from the H2 data sheet and [24]. A single scalar η multiplies p_1, p_2, e_{01}, e_{10} and divides T_1, T_2 , so $\eta = 1$ is the base device; the four channels therefore co-vary in the fixed ratio the profile sets, and every threshold is quoted both as η^* and in device units. Per-channel sweeps in which only one family is switched on separate their contributions.

3.3 What is measured, where

The interferometer contributes two circuits, one for $\text{Re}\langle V \rangle$ and one (with an $S^\dagger$ on the ancilla) for $\text{Im}\langle V \rangle$. The φ_G integrands are measured at $M = 48$ nodes per leg, placed on a mirror-symmetric grid in λ so that the leg-to-leg cancellation of the dynamical phase is exact at the level of the quadrature. At each node the Pauli strings of $\hat{\mathcal{H}}$ are partitioned into qubit-wise-commuting groups, each measured by its own circuit with its own shot record. For the overlaps, generators inside one layer have disjoint support and therefore commute both with each other and with each other’s basis rotations, so $\langle \hat{A}_\mu \rangle$ is unchanged by them: the prefix circuit of every generator in layer ℓ is “layers $1 \dots \ell - 1$ plus a basis rotation”. One noisy state is produced per layer, and each generator then draws its *own independent* shot record from it — the shot noise is that of one circuit execution per parameter, as it would be on hardware.

The classical driver holds $\theta(t)$ on the trajectory’s own RK4 grid (6×10^2 to 2×10^3 steps) and interpolates the measured integrands onto it before performing the quadratures. Simulating a noisy density matrix at every RK4 step is out of reach, so we simulate the measurement at the M nodes and transport it by writing each integrand as (noiseless value) + (measured deviation), integrating the

first part on the full RK4 grid and the interpolated second part alongside it. The deviation vanishes identically at zero noise and infinite shots, so the estimator carries *no* classical quadrature bias into the threshold; the residual is the interpolation error of the noise-induced deviation. Repeating the noiseless evaluation at $M \in \{12, 24, 48, 96\}$ confirms this: the estimator returns the trajectory’s own algorithmic deviation ($\leq 3.1 \times 10^{-4}$ for the non-interacting and $\leq 1.8 \times 10^{-3}$ for the interacting non-trivial configuration) with no systematic drift in M (`logs/quadrature_check.json`), and, where the noisy deviation is above the shot floor, it changes by less than 5% between $M = 24$ and $M = 48$ (`logs/node_conv/`).

3.4 Phase estimator and branch policy

A single real-part Hadamard test estimates φ_{qc} through $P_{|0\rangle} = \frac{1}{2}(1 + \cos \varphi_{\text{qc}})$, whose slope vanishes exactly at the quantised targets 0 and π . We therefore measure both quadratures and use

$$\varphi_{\text{qc}} = \text{atan2}(2\hat{P}_{|0\rangle}^{\text{im}} - 1, 2\hat{P}_{|0\rangle}^{\text{re}} - 1). \quad (3)$$

This is well conditioned everywhere on the circle, and it has a second property that turns out to matter more: any channel that multiplies both quadratures by a common factor — global depolarisation, dephasing, a symmetric readout error — cancels identically in the ratio, so such noise costs φ_{qc} variance but no bias.

φ_G is accumulated continuously without wrapping, so $\varphi_B = \varphi_{\text{qc}} + \varphi_G$ is reported both raw (an odd multiple of π for $U = 10$) and as its principal value. Every deviation and every discrimination margin is computed *on the circle*, $|\varphi_B - \varphi_B^{\text{ED}}|_{\text{circ}} = |((\varphi_B - \varphi_B^{\text{ED}} + \pi) \bmod 2\pi) - \pi|$, so estimates straddling the $\pm\pi$ branch boundary — which is exactly where the non-trivial target sits — are compared correctly.

3.5 Accuracy criterion, threshold rule, and reference

The reference φ_B^{ED} is computed at runtime by our own pipeline as the gauge-invariant Wilson loop $-\text{Im} \ln \prod_k \langle G(\lambda_k) | G(\lambda_{k+1}) \rangle$ over 400 exactly diagonalised instantaneous ground states; it returns 0 and π to 10^{-8} in all configurations. Nothing is hard-coded.

We fix the tolerance $\epsilon_{\text{tol}} = 0.1\pi \approx 0.3142$ rad: ten per cent of the quantisation gap, which guarantees a discrimination margin of at least 0.8π between the two topological phases and is loose enough that the threshold is set by device physics rather than by the residual algorithmic error of the noiseless anchor ($\lesssim 10^{-3}$). A stricter $\epsilon_{\text{tol}} = 0.0628$ rad is reported as a sensitivity check. The reference shot count is $S_{\text{ref}} = 10^4$ per circuit and every point uses 32 independent seeds.

Threshold rule. For each configuration, let $\bar{\Delta}(\eta)$ be the seed-median circle deviation on the logarithmic η grid. Let $\eta_{<}$ be the largest grid value at which $\bar{\Delta} \leq \epsilon_{\text{tol}}$ and at which the same holds for every smaller grid value (sustained compliance — this prevents a lucky high- η point from being read as a threshold). η^* is the log-linear interpolation of $\bar{\Delta}$ to ϵ_{tol} between $\eta_{<}$ and the next grid point — linear in $\bar{\Delta}$, logarithmic in η . Uncertainties are the 16–84 percentile band of 2000 bootstrap resamples over the seeds. The same rule is applied unchanged to every configuration, every shot count, every channel subset, both profiles, and the mitigated and Trotterised runs.

Interpolating logarithmically in *both* variables instead is an equally defensible reading of the same rule, and it is worth knowing that nothing here turns on the choice: it moves the four primary thresholds to 0.306, 0.14, 0.00397 and 0.00449, a shift of at most 4%, which leaves 3 of the 4 inside the bootstrap band of the quoted value (the exception, $U = 0$ trivial, misses its band edge by less than a part in a hundred). Both conventions are computed for every configuration in `logs/analysis.json`; every number quoted in this paper uses the first. The bracket $[\eta_{<}, \eta_{<+1}]$, and hence the compliance test that does the real work, is identical either way — the grid spacing of $\approx \sqrt[4]{10}$, not the interpolant, sets the resolution.

Shot floor. Because finite sampling persists even for a fault-tolerant device, we report it separately: S^* is the smallest grid shot count whose seed-median deviation meets the tolerance and stays compliant for all larger counts, evaluated at a fixed low incoherent noise $\eta = 10^{-3}$.

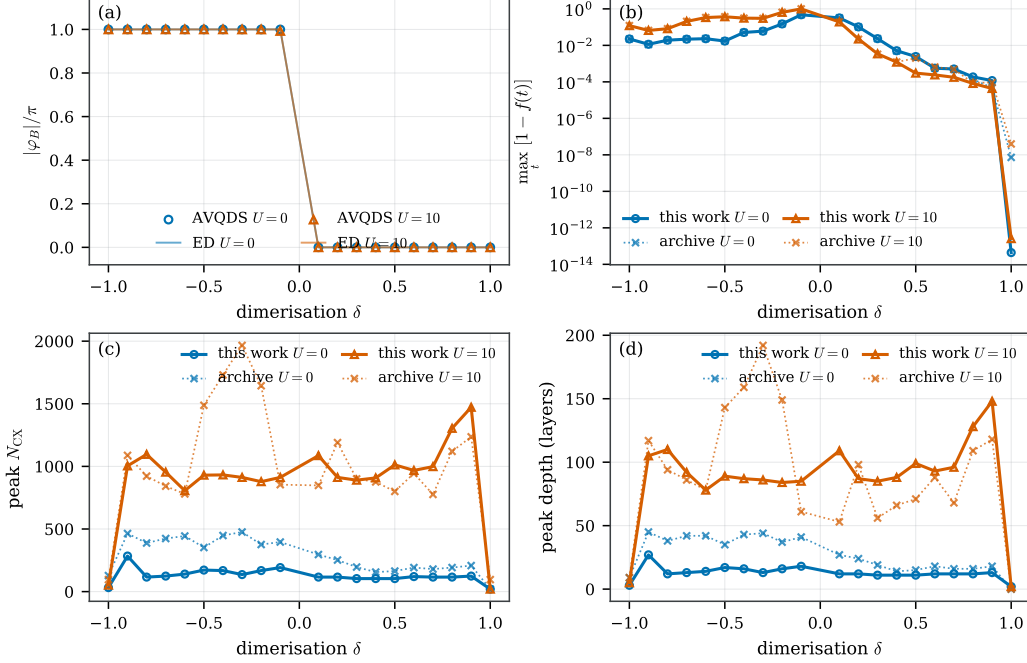


Figure 1: Noiseless re-implementation across the topological transition at $T = 20$, compared with exact diagonalisation and with the published traces of the original study. (a) The computed Berry phase is quantised at 0 for $\delta > 0$ and π for $\delta < 0$ at all twenty dimerisations and both interaction strengths. (b) The loop-resolved infidelity with the instantaneous ground state — a property of the physical trajectory rather than of the particular ansatz — agrees with the published traces to three digits, including the sharp rise on approaching the transition. (c, d) CNOT count and circuit depth. Our adaptive growth reaches the same accuracy with a more compact ansatz than the published runs. The interacting-over-non-interacting ratio that drives the depth dependence of the noise study survives (6.6 in layers, against 4.4 published); the excess cost of the non-trivial over the trivial region does not (Sec. 4).

4 The noiseless anchor

We re-grew every ansatz from scratch and swept the whole dimerisation axis, $\delta \in \{\pm 0.1, \dots, \pm 1.0\}$, for both $U = 0$ and $U = 10$ at $T = 20$ (Fig. 1). The folded Berry phase is 0 for $\delta > 0$ and π for $\delta < 0$ at all forty points. At the four configurations used for the noise study the circle deviations from exact diagonalisation are 7.7×10^{-5} and 2.6×10^{-5} ($U = 0$, non-trivial/trivial) and 1.1×10^{-3} and 1.6×10^{-4} ($U = 10$), i.e. at or below the $\sim 10^{-3}$ accuracy the method is expected to reach.

Beneath these numbers sits a unit-test suite that compares every primitive with a brute-force construction: the Jordan–Wigner Hamiltonian against an explicit second-quantised build with dense fermion operators (exact agreement), the analytic $|\partial_\mu \Psi\rangle$ against central differences (7×10^{-11}), M , V and L^2 against dense density-matrix traces ($< 10^{-9}$), every noise channel against its explicit Kraus sum, and the scheduled circuit at zero rates against the exact state vector (9×10^{-16}). The variational dynamics itself is checked against exact propagation: over the forward half of the loop the AVQDS state stays within 1.3×10^{-4} ($U = 0$) and 2.7×10^{-4} ($U = 10$) of the exactly evolved state (logs/regularization_scan.json), the same order as the $\sim 10^{-4}$ the original study reports for its own runs, so the residual mid-loop infidelity in Fig. 2(b) is non-adiabaticity and not variational error. Three further checks establish that we are simulating the same physical object as the original study. First, an *exact* (non-variational) time-reversed loop propagated with the same schedule returns the Wilson-loop Berry phase to 5×10^{-14} in the worst of 4 cases (both topological regimes $\times$ two loop times at $U = 0$; logs/validation_berry.json), which validates the Hamiltonian, the twist convention, the loop and the dynamical-phase cancellation independently of anything variational (test_berry.py). Second, the AVQITE ground-state circuit for $U = 0$ contains exactly 6 generators and 20 CNOTs, matching the published gate census (four weight-two and two weight-four generators, 20 CNOTs) at every δ . Third, and most stringently, the loop-resolved infidelity — a property of the

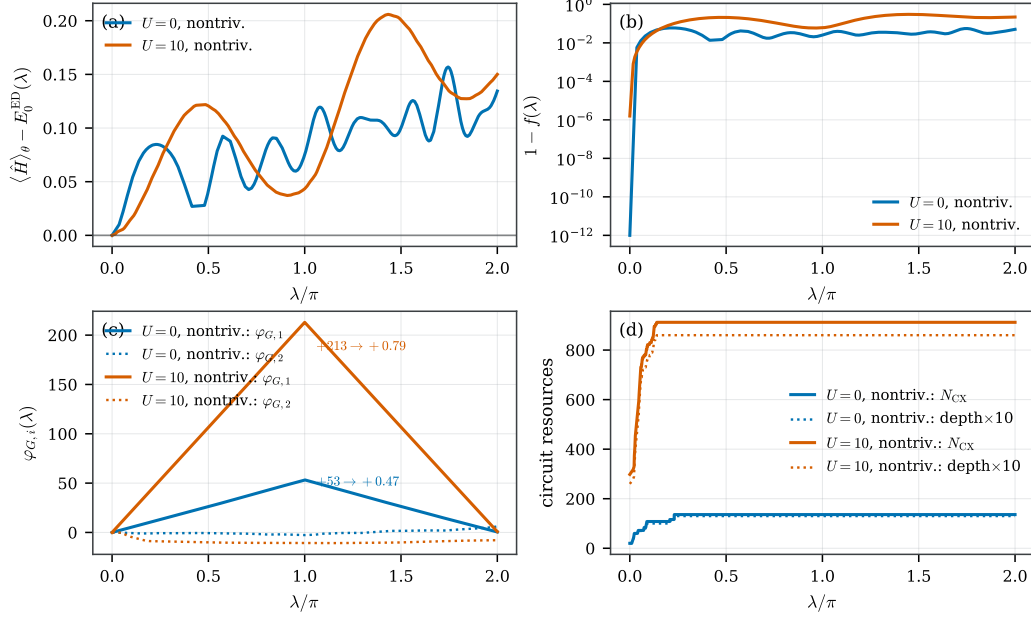


Figure 2: Loop-resolved behaviour for the two non-trivial configurations. (a) Variational energy against the instantaneous exact ground-state energy. (b) Infidelity with the instantaneous ground state; at $T = 20$ the evolution is strongly non-adiabatic, and the Berry phase is nevertheless correct. (c) The two global-phase legs. $\varphi_{G,1}$ is the residue of two half-loop integrals of 53 and 213 rad that cancel by time reversal down to $+0.47$ and $+0.79$; $\varphi_{G,2}$ is the residue of the ansatz-growth asymmetry between the legs. (d) Growth of the circuit resources along the loop; almost all growth happens in the first half.

physical trajectory, not of the ansatz — agrees with the published traces to three digits: at $\delta = -0.3$ we obtain 6.00×10^{-2} against 6.025×10^{-2} , and at $\delta = -0.1$ we obtain 4.760×10^{-1} against 4.765×10^{-1} (both read from `logs/analysis.json` and `logs/archive_reference.json`). The leg $\varphi_{G,1}$ at the loop end likewise agrees, $+0.472$ against the published $+0.470$.

Our adaptive growth is more economical than the published runs. At $(U, \delta) = (0, -0.3)$ we reach a peak depth of 13 layers and 136 CNOTs with 21 parameters, where the published trace has 44 layers and 476 CNOTs with 68 parameters, at a *larger* phase error; at $(10, -0.3)$ we use 86 layers and 912 CNOTs against 192 and 1966. The likely origin is the L^2 estimate discussed in Sec. 2: a regularised solve that inflates L^2 triggers extra growth. Two consequences must be stated plainly. First, a shallower circuit is a more favourable noise environment, so every threshold below should be read as an upper bound on what the published circuits would achieve. Second, the resource *ratios* do not all survive: the interacting circuits remain several times deeper than the non-interacting ones (86 against 13 layers, a factor 6.6, against 4.4 in the published traces), but the excess cost of the non-trivial over the trivial region, a factor 2.4 ($U = 0$) and 2.2 ($U = 10$) in the published traces at $\delta = \pm 0.3$, is reduced to 1.3 and 1.0 in ours. The physical asymmetry between the two regimes is intact — the peak infidelity differs by a factor 2.6 at $U = 0$ and 88 at $U = 10$, against 2.6 and 92 in the published traces — but our growth heuristic does not translate it into gate count in the same way. We use the same (δ, T) as the published benchmark point and report both sets of resources.

5 Noise resilience and thresholds

5.1 Where the method breaks

Figure 3(a) shows every point of the main sweep: four configurations $\times$ eighteen values of η including the noiseless-channel limit, or 72 points, each carrying eight shot counts and 32 seeds, with nothing withheld. Counting the mitigated, trapped-ion and per-channel sweeps as well, 214 points were run in total, and the per-seed values of φ_{qc} , $\varphi_{G,1}$, $\varphi_{G,2}$ and φ_B for every one of them are in `logs/`. The two interaction strengths behave qualitatively differently. For the shallow non-interacting circuits the

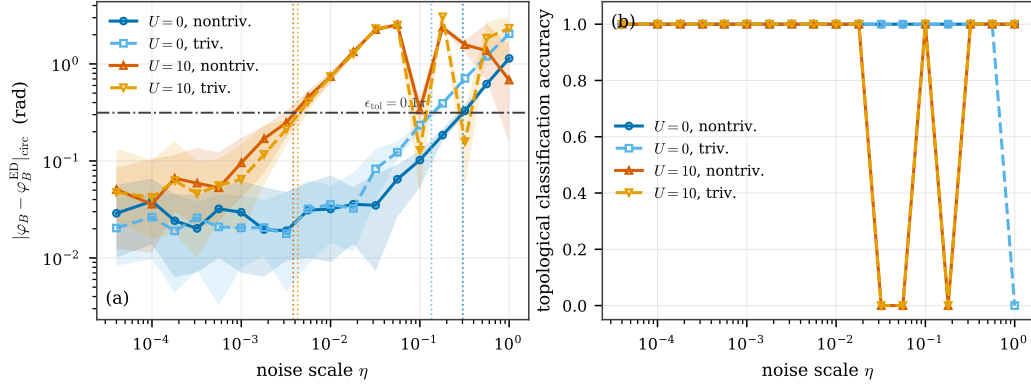


Figure 3: (a) Circle deviation of the Berry phase from the exact reference against the noise scale η at $S_{\text{ref}} = 10^4$ shots per circuit; lines are seed medians, bands the 16–84 percentile over 32 seeds, the dash-dotted line the tolerance and the dotted verticals the extracted thresholds. The leftmost point is the noiseless-channel limit, where only finite sampling remains. (b) Accuracy of the nearest-quantised-value classification of the topological phase.

Table 1: Berry phase, accuracy and extracted noise threshold for the four configurations at $T = 20$, with $S_{\text{ref}} = 10^4$ shots per circuit and 32 seeds. Berry phases are principal values in units of π (exact reference 0 for $\delta > 0$, π for $\delta < 0$). Deviations are computed on the circle; the noiseless-anchor row gives the algorithmic deviation, the noisy rows the seed median $\pm$ half the 16–84 percentile spread. The threshold is the noise scale η^* with its bootstrap band and the corresponding two-qubit error rate on profile SC-A; Table 2 gives the full device-referenced translation. Depth and N_{CX} are the peak values of the variational loop circuit.

Configuration	Noise setting	φ_B/π	$ \varphi_B - \varphi_B^{\text{ED}} _{\text{circ}}$	Threshold	Depth	N_{CX}
$U=0, \delta=-0.3$	noiseless anchor	+1.0000	7.7×10^{-5}			
	$\eta^*/10$ ($\eta=0.018$)	+0.9841	0.036 ± 0.020	$0.30^{+0.01}_{-0.008}$	13	136
	just below η^* ($\eta=0.18$)	+0.9412	0.185 ± 0.027	$\epsilon_{2q}=6.1 \times 10^{-4}$		
	above $3\eta^*$ ($\eta=1$)	+0.6370	1.141 ± 0.055			
$U=0, \delta=+0.3$	noiseless anchor	+0.0000	2.6×10^{-5}			
	$\eta^*/10$ ($\eta=0.01$)	-0.0066	0.036 ± 0.021	$0.13^{+0.004}_{-0.003}$	11	104
	just below η^* ($\eta=0.1$)	-0.0745	0.234 ± 0.029	$\epsilon_{2q}=2.7 \times 10^{-4}$		
	above $3\eta^*$ ($\eta=0.56$)	-0.3823	1.201 ± 0.032			
$U=10, \delta=-0.3$	noiseless anchor	-0.9996	1.1×10^{-3}			
	$\eta^*/10$ ($\eta=0.00032$)	-0.9532	0.059 ± 0.058	$0.0038^{+0.0002}_{-0.0002}$	86	912
	just below η^* ($\eta=0.0032$)	-0.9209	0.249 ± 0.077	$\epsilon_{2q}=7.6 \times 10^{-6}$		
	above $3\eta^*$ ($\eta=0.018$)	-0.5764	1.331 ± 0.083			
$U=10, \delta=+0.3$	noiseless anchor	-0.0001	1.6×10^{-4}			
	$\eta^*/10$ ($\eta=0.00032$)	+0.0114	0.046 ± 0.045	$0.0043^{+0.0003}_{-0.00005}$	85	890
	just below η^* ($\eta=0.0032$)	+0.0676	0.212 ± 0.059	$\epsilon_{2q}=8.6 \times 10^{-6}$		
	above $3\eta^*$ ($\eta=0.018$)	+0.4047	1.271 ± 0.096			

deviation is *flat* over more than two decades of η , at the finite-shot floor of $2\text{--}3 \times 10^{-2}$ rad, and only leaves it above $\eta \approx 3 \times 10^{-2}$; adding incoherent noise at the base-device ratio has no measurable effect on the Berry phase until the rates approach a few per cent of a real transmon. For the deep interacting circuits there is no such plateau; the curve leaves its floor of $\approx 5 \times 10^{-2}$ rad already at $\eta \approx 10^{-3}$ and rises steadily.

The thresholds are collected in Tables 1 and 2, and the depth dependence is the headline. With 136 CNOTs and 13 layers, the non-interacting non-trivial configuration tolerates $\eta^* = 0.303$ (band 0.294–0.315) — a two-qubit error rate of $\epsilon_{2q} = 6.1 \times 10^{-4}$, an ancilla readout error of 3.9×10^{-3} and $T_1 = 0.66$ ms, i.e. a device about 3 times better than the base profile in every rate simultaneously. With the 912-CNOT, 86-layer interacting circuits the threshold falls by nearly two orders of magnitude, to $\eta^* = 0.00382$ (0.00364–0.00405), $\epsilon_{2q} = 7.6 \times 10^{-6}$, $T_1 = 52$ ms: early-fault-tolerant territory,

Table 2: Thresholds in device-referenced units. Each row translates η^* of Table 1 into the four rates of profile SC-A, which co-vary along η in the ratio the profile fixes. S^* is the separately extracted shot floor at fixed low incoherent noise ($\eta = 10^{-3}$), at the primary and at the stricter tolerance. The last two columns give the threshold with the mitigation of Sec. 8, on the trapped-ion-class profile, and for the deep-circuit Trotterised baseline at the same accuracy target.

Configuration	η^*	ϵ_{2q}^*	e_{10}^*	T_1^* [ms]	$S_{0.1\pi}^*$	$S_{0.02\pi}^*$	η_{mit}^*	$\eta_{\text{TI-B}}^*$	η_{Trotter}^*
$U=0, \delta=-0.3$	$0.30^{+0.01}_{-0.008}$	6.1×10^{-4}	3.9×10^{-3}	0.66	1×10^2	3×10^3	0.63	1.3	0.010
$U=0, \delta=+0.3$	$0.13^{+0.004}_{-0.003}$	2.7×10^{-4}	1.8×10^{-3}	1.5	1×10^2	3×10^3	0.23	0.38	0.010
$U=10, \delta=-0.3$	$0.0038^{+0.0002}_{-0.0002}$	7.6×10^{-6}	5.0×10^{-5}	52	1×10^3	$> 10^6$	0.0056	0.010	0.0067
$U=10, \delta=+0.3$	$0.0043^{+0.0003}_{-0.00005}$	8.6×10^{-6}	5.6×10^{-5}	47	3×10^2	$> 10^6$	0.0080	0.0095	0.0073

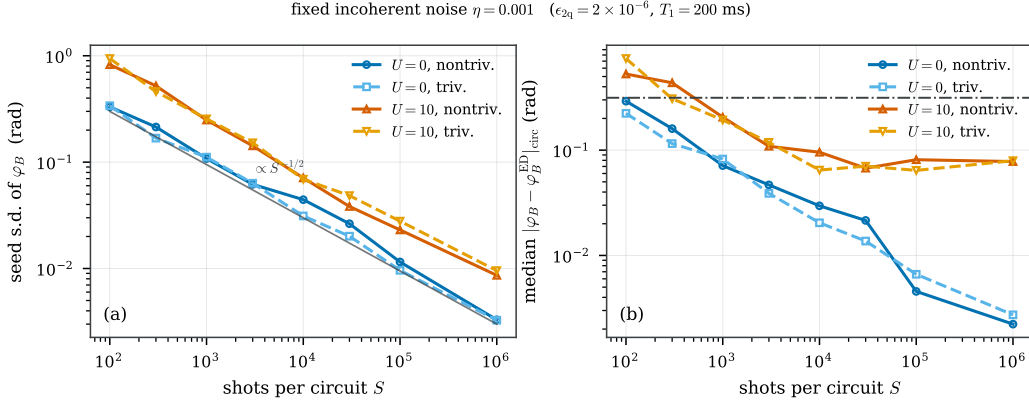


Figure 4: Finite sampling as an independent axis, at fixed low incoherent noise $\eta = 10^{-3}$. (a) Seed standard deviation of the Berry phase against the number of shots per circuit; the grey guide is $S^{-1/2}$. (b) Median circle deviation with the tolerance; the crossings define the shot floor S^* .

some 262 times better than the base profile. At the stricter tolerance the same ordering holds with every threshold reduced by a further factor 5–23 ($\eta^* = 0.054$ and 0.000169 for the two non-trivial configurations).

The trivial and non-trivial configurations at the same U do not share a threshold, and neither ordering is universal: at $U = 0$ the non-trivial regime is the more robust (0.303 against 0.135), at $U = 10$ the trivial one is (0.00429 against 0.00382). The difference tracks the size of the noiseless φ_G that each configuration has to reconstruct rather than any topological property; we report all four and use the more conservative of each pair below.

5.2 Distinguishing the topological phases

Below threshold the two regimes stay separated. The circle distance between the trivial and non-trivial estimates never falls below 0.958π ($U = 0$) and 0.983π ($U = 10$) over the compliant range, and the nearest-quantised-value classification (Fig. 3(b)) is perfect for every seed. Taking the more conservative of the two same- U thresholds gives $\eta^* = 0.135$ ($\epsilon_{2q} = 2.7 \times 10^{-4}$) for the non-interacting chain and $\eta^* = 0.00382$ ($\epsilon_{2q} = 7.6 \times 10^{-6}$) for the interacting one.

Classification survives further than phase accuracy does, and the gap is instructive: it stays perfect up to $\eta = 0.56$ and $\eta = 0.018$ respectively, well beyond the accuracy thresholds. Deciding *which* of two quantised values a phase is closer to tolerates an error of nearly $\pi/2$, whereas reporting the phase itself to 0.1π does not. A study that only asked “does the algorithm still see the transition?” would therefore have reported a threshold an order of magnitude more optimistic than the accuracy threshold; both numbers are given here.

5.3 The shot floor

Finite sampling is not a technology problem: the final logical measurement of a fault-tolerant device is still a projective sample, so this axis survives error correction. Figure 4 isolates it at a fixed low incoherent noise. The seed standard deviation follows $S^{-1/2}$ cleanly over four decades, which confirms that the sampling model is doing what it should, and the prefactor is what distinguishes the configurations: it is set by $\sigma_{\varphi_{G,1}} \simeq \sigma_E T / \sqrt{2M}$, so it grows with the loop time and with the norm of the Hamiltonian, and the interacting chain — whose $\|\hat{\mathcal{H}}\|$ is four times larger — pays for it.

At the primary tolerance the floors are modest: $S^* = 10^2$ shots per circuit for the non-interacting configurations and 10^3 for the interacting non-trivial one (Table 2). Sampling is therefore *not* the binding constraint at 0.1π ; the incoherent channels are, by one to two orders of magnitude in η . It becomes binding as soon as more is asked of the estimate: at the stricter $\epsilon_{\text{tol}} = 0.0628$ rad the floor rises to 3×10^3 for the non-interacting chain, and the interacting chain never meets it at any shot count we swept, because its curve saturates at ≈ 0.08 rad — a *bias* left by $\eta = 10^{-3}$, which no number of shots can remove. That saturation, visible in Fig. 4(b), is the cleanest single demonstration that the two axes are physically distinct.

Two consequences are worth stating. First, the shot floor and the incoherent threshold are genuinely independent axes: adding shots does nothing about the $\varphi_{G,1}$ bias of Sec. 6, and reducing ϵ_{2q} does nothing about the sampling variance. Second, because the number of measured circuits per Berry-phase estimate is $2 + 2(M+1)(N_\theta + n_g)$ (Sec. 9) and S^* multiplies it, the total shot cost of one topological invariant on this eight-qubit chain is already in the 10^7 – 10^8 range.

6 Mechanism

6.1 Why the interferometric leg is not the problem

Let $\mathcal{E}$ be any channel that acts on the interferometer as a common contraction of both quadratures, $\langle V \rangle \mapsto F \langle V \rangle$ with $F \in (0, 1]$ real. Global depolarisation does this exactly; dephasing and a *symmetric* readout error do it to the accuracy with which they are symmetric. The single real-part estimator $\hat{\varphi}_{\text{qc}} = \arccos(2\hat{P}_{|0\rangle}^{\text{re}} - 1)$ then acquires the bias $\arccos(F \cos \varphi_{\text{qc}}) - \varphi_{\text{qc}}$, which is large — it pushes every phase towards $\pi/2$ — and is worst exactly at the quantised targets, where the slope $\partial P / \partial \varphi_{\text{qc}}$ vanishes. The two-quadrature estimator of Eq. (3) removes it identically: $\text{atan2}(Fy, Fx) = \text{atan2}(y, x)$. What survives is variance, not bias: the shot noise on each quadrature is $O(S^{-1/2})$ irrespective of F , but its effect on the angle is amplified by $1/F$, so $\sigma_{\varphi_{\text{qc}}} \simeq 1/(F\sqrt{S})$. The measured contrast $F = |2P_{|0\rangle}^{\text{re}} - 1 + i(2P_{|0\rangle}^{\text{im}} - 1)|$ is logged at every sweep point and is what sets that amplification.

This is visible directly in Fig. 5(a, b): over the whole compliant range the φ_{qc} bias stays far below the tolerance while its standard deviation tracks $1/F$, and the two curves cross only where the contrast has already collapsed. Quantitatively, evaluating both estimators on the *same* noisy output distributions at each configuration's threshold, the two-quadrature bias is $< 10^{-3}$ and $< 10^{-3}$ rad (non-interacting and interacting non-trivial) while the single-quadrature $\arccos$ estimator biases by 0.575 and 0.130 rad at the same contrast ($F = 0.706$ and 0.861). Taking the largest η at which each estimator keeps $|\Delta\varphi_{\text{qc}}| \leq \epsilon_{\text{tol}}$, the two-quadrature readout survives to $\eta = 1$ and 0.56 against 0.1 and 0.032 — a factor 10–20 from the estimator alone, before any noise is removed from the device. The single-quadrature readout also cannot resolve the sign of φ_{qc} at all, which the raw φ_B of the interacting configuration (-2.9996π) needs.

6.2 Why the dynamical leg is

$\varphi_{G,1} = -\int \langle \hat{\mathcal{H}} \rangle_\theta dt$ is a different animal. Along the forward half of the loop it accumulates $\int_0^{T/2} |\langle \hat{\mathcal{H}} \rangle| dt \approx 53$ rad for $U = 0$ and ≈ 213 rad for $U = 10$ (Fig. 2(c)), and the return half, propagated with $dt < 0$ through the mirror-image λ values, cancels it down to a residue of $+0.47$ and $+0.79$ rad respectively — amplification factors of 113 and 270. The cancellation is exact only if the measured energy obeys $\langle \hat{\mathcal{H}} \rangle(\lambda) = \langle \hat{\mathcal{H}} \rangle(2\pi - \lambda)$.

Noise breaks that symmetry in a specific, unavoidable way. A depolarising channel contracts every traceless observable, $\langle \hat{\mathcal{H}} \rangle \mapsto c_{\mathbb{I}} + F(\lambda) [\langle \hat{\mathcal{H}} \rangle - c_{\mathbb{I}}]$, with a survival factor $F(\lambda)$ set by the number of gates executed so far. But the adaptive ansatz *grows monotonically* along the loop, and almost all of that growth happens in the first half (Fig. 2(d)): the circuit that measures $\langle \hat{\mathcal{H}} \rangle$ at λ and the one that measures it at $2\pi - \lambda$ have the same generator count only after growth has saturated. The two halves are therefore contracted by different factors, and what reaches φ_B is

$$\delta\varphi_{G,1} \simeq [F^+ - F^-] \times \left| \int_0^{T/2} \langle \hat{\mathcal{H}} \rangle dt \right|, \quad (4)$$

i.e. a *relative* asymmetry of 10^{-3} in the measured energy becomes a phase error of order 0.05 rad for $U = 0$ and 0.2 rad for $U = 10$. The same factor multiplies the shot noise: with quadrature weights of order $T/(2M)$ and an energy sampling error σ_E , the accumulated fluctuation is $\sigma_{\varphi_{G,1}} \simeq \sigma_E T / \sqrt{2M}$, growing linearly with the loop time and with $\|\hat{\mathcal{H}}\|$.

Measuring each leg separately makes the consequence explicit. Taking the largest η at which each leg's own bias stays within ϵ_{tol} , the interacting non-trivial configuration gives $\eta = 0.0032$ for $\varphi_{G,1}$, 0.018 for $\varphi_{G,2}$ and 0.32 for φ_{qc} , against a whole-method threshold of 0.00382: the dynamical leg alone accounts for the failure, going first by a factor of 6 over the geometric leg and 100 over the interferometer.

The shallow configuration tells a more interesting story. There the individual legs cross at 0.18 ($\varphi_{G,1}$) and 0.1 ($\varphi_{G,2}$) while the whole method survives to 0.303 — the assembled Berry phase is *more* accurate than either of its geometric-plus-dynamical parts. This is the noiseless cancellation argument surviving into the noisy regime, but only partially and only for the deterministic biases: φ_{qc} and $\varphi_{G,2}$ are bounded quantities read out at the end of a closed loop and they share the frozen trajectory, so their systematic distortions are correlated and partly subtract. What does not subtract is the stochastic part, drawn independently on each of the three circuit families, and what has no partner at all is $\varphi_{G,1}$ — an unbounded time integral whose smallness is an accident of time-reversal symmetry rather than of the geometry. As soon as the circuits are deep enough for the amplified $\varphi_{G,1}$ drift to exceed the cancellation the legs can supply, which happens between $U = 0$ and $U = 10$ here, the whole-method threshold collapses onto the dynamical leg's.

6.3 Which channel, and which circuit property

The per-channel sweeps of Fig. 5(c, d) switch on one family at a time, and the answer depends on the circuit — which is itself the point.

For the shallow non-interacting circuits (136 CNOTs, 13 layers) *no gate channel reaches the tolerance anywhere in the swept range*: alone, two-qubit depolarising, T_1/T_2 and single-qubit depolarising leave deviations of 0.11, 0.17 and 0.05 rad even at $\eta = 1$, i.e. at the full base-device rates. The only family that crosses on its own is readout error, at $\eta = 0.379$ — close to the all-channel threshold 0.303 — and it is the one channel whose magnitude does *not* scale with the circuit.

For the deep interacting circuits (912 CNOTs, 86 layers) the ordering inverts completely: T_1/T_2 relaxation crosses first ($\eta = 0.00666$), two-qubit depolarising next (0.0105), single-qubit depolarising much later (0.0799), and readout never crosses at all (it leaves 0.28 rad at $\eta = 1$, against 2.26 for two-qubit depolarising). Relaxation leads because the loop's wall-clock duration is set by the number of *layers*, which grows by a factor 6.6 between the two interaction strengths, while the CNOT count grows by 6.7.

Two structural remarks explain the readout behaviour. First, the two-quadrature estimator is blind to the symmetric part of the readout channel (Sec. 6), so only the $e_{01} \neq e_{10}$ asymmetry and the weight-dependent contraction of the measured Pauli parities survive; both act through φ_G , not through φ_{qc} . Second, readout error damages $\varphi_{G,1}$ only through the *traceless* part of $\hat{\mathcal{H}}$: the identity component — $-UN/4 = -10$ of the $U = 10$ energy — is a classically known constant that never passes through a measurement and cancels exactly between the legs. The interacting chain therefore exposes a smaller fraction of its (much larger) half-loop dynamical phase to readout error than the non-interacting one does, which is why readout matters less there in absolute terms despite the larger amplification factor.

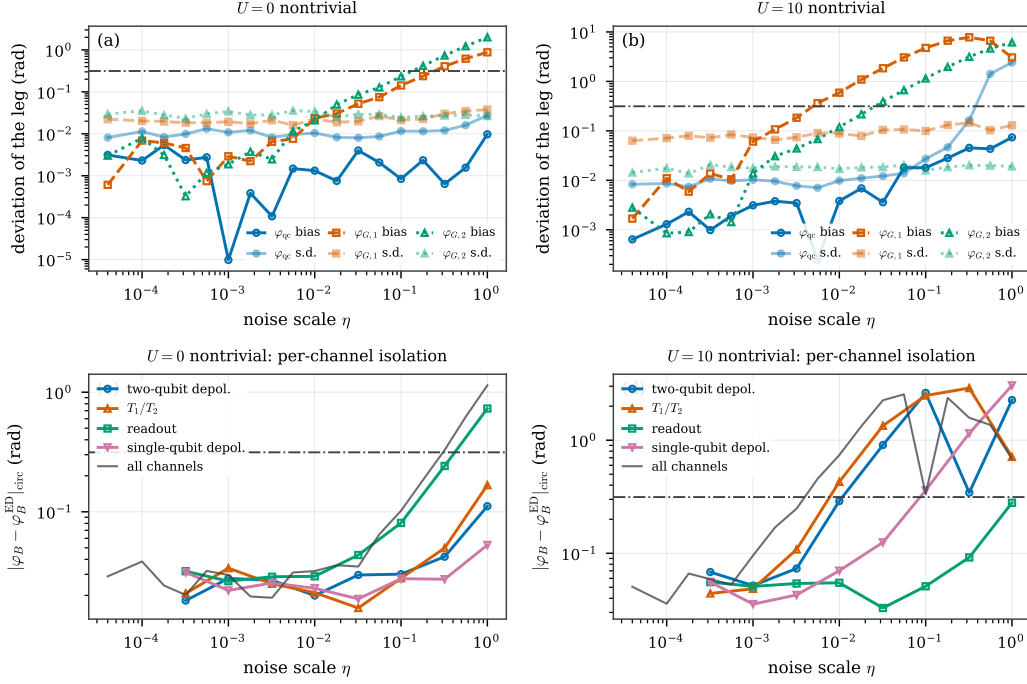


Figure 5: (a, b) Seed-median absolute deviation (solid markers) and seed standard deviation (faded) of each of the three legs of $\varphi_B = \varphi_{qc} + \varphi_{G,1} + \varphi_{G,2}$, relative to its noiseless-channel value, for the non-interacting and interacting non-trivial configurations. The median is taken rather than the mean because φ_{qc} lives on the circle: once the interferometer decoheres its residuals spread over the whole circle and would cancel in a mean. (c, d) Per-channel isolation: the same sweep with only one noise family switched on.

6.4 Depth dependence

The threshold ratio between the interacting and non-interacting configurations tracks the circuit cost rather than any property of the physics: the interacting circuits carry 912 CNOTs against 136 and 86 layers against 13. Figure 6(b) makes the point in a channel-independent way by plotting the phase error against the *output-state infidelity* rather than against η : the two families of curves do not collapse. At their respective thresholds the shallow circuits have already lost a fraction 0.175 of their fidelity while the deep ones fail at 0.015 — an order of magnitude less state damage for the same phase error. Equal damage to the state does not produce equal damage to the Berry phase, because what matters is not how far the state has moved but how asymmetrically the measured energy has been contracted between the two halves of the loop, and that asymmetry is itself a function of how much the ansatz grew. It also means that the output fidelity, the natural figure of merit for a state preparation, is a poor proxy for the accuracy of this observable in either direction.

7 Failure analysis

Three distinct failure modes appear as η grows, and they fail differently.

Amplified drift of the dynamical leg (first to bite). Well before the state is visibly damaged — at output infidelities of a few times 10^{-2} — $\varphi_{G,1}$ has already drifted past the tolerance for the deep circuits, by the mechanism of Eq. (4). This failure is a *bias*: it does not average away over seeds, and adding shots does not help. It is the mode that sets the threshold in Table 1.

Contrast collapse of the interferometer (second). As $F \rightarrow 0$ the two-quadrature estimator remains unbiased but its variance diverges as $1/(F\sqrt{S})$; once F falls to a few per cent the estimated φ_{qc} is essentially uniform on the circle and the reported Berry phase is a random angle. This failure is a *variance*: it is visible as the widening of the seed bands in Fig. 3(a) and it is the mode that

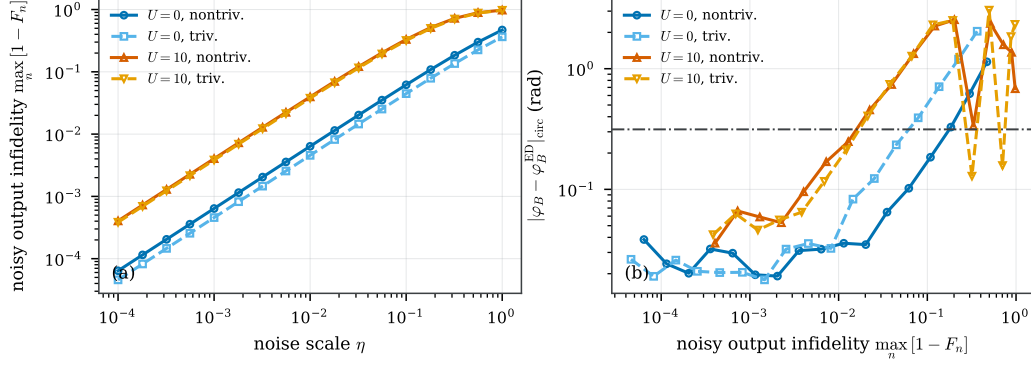


Figure 6: (a) Infidelity of the noisy output state with the ideal state along the loop against η . (b) Phase error against infidelity: the same state-level damage produces very different phase errors in the shallow and deep circuits.

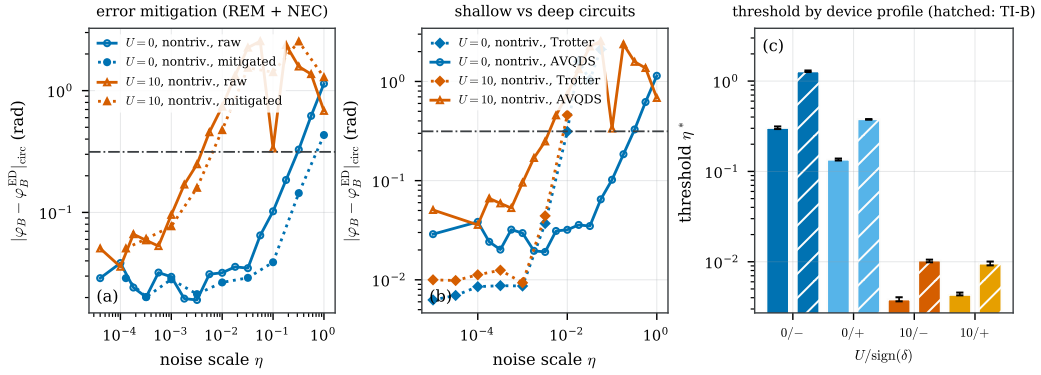


Figure 7: (a) Error mitigation: tensored readout unfolding plus a noise-estimation-circuit rescaling of the φ_G integrands, against the raw estimate. (b) The shallow adaptive loop against a first-order Trotterised loop of the same task, both within 10^{-3} of the exact value noiselessly. (c) Extracted thresholds on the superconducting-class (SC-A) and trapped-ion-class (TI-B) profiles.

finally destroys the topological classification. It can be delayed by shots, but only as $S^{1/2}$ against an exponential collapse of F with depth, which is why more shots buy so little at large η .

Saturation, and why the error curve is not monotone. At the largest η the deviation stops growing and settles near the mean circle distance between a uniformly random angle and the target. The median deviation can therefore *decrease* between two large- η points purely by chance, which is exactly why the pre-registered threshold rule demands sustained compliance rather than the largest complying η : without it, a lucky saturated point would be read as a threshold two decades above the true one.

What does not fail. The branch structure never causes a spurious failure. For $U = 10$ the raw phase is an odd multiple of π (-2.9996π noiselessly), so the folded value sits exactly on the $\pm\pi$ branch cut and an arbitrarily small perturbation flips its sign; because every deviation and every margin is computed on the circle, this shows up as a deviation of $O(10^{-3})$ rather than of 2π . Reporting principal values and comparing them linearly would have manufactured a spurious catastrophic failure here.

8 Extensions

8.1 Mitigation aimed at the dominant leg

Because the damage is concentrated in $\varphi_{G,1}$, and because that damage is a contraction of a measured expectation value, the natural countermeasure is a noise-estimation circuit [25]: the same circuit with every rotation angle set to zero has the known output $|\Psi_0\rangle$, its survival probability gives the depolarising contrast F at that point of the loop, and $\langle\hat{H}\rangle$ and the a_μ are rescaled by $1/F$. We combine it with tensored readout unfolding [26, 27] using the known confusion matrix. No mitigation is applied to φ_{qc} , which does not need it. The threshold rises in every configuration — by $2.07\times$ and $1.70\times$ at $U = 0$ and $1.47\times$ and $1.87\times$ at $U = 10$ (Fig. 7(a), Table 2) — taking the interacting chain from $\epsilon_{2q} = 7.6 \times 10^{-6}$ to $\epsilon_{2q} = 1.1 \times 10^{-5}$. The gain is real but modest, and its size is itself diagnostic: a rescaling can only remove the part of the damage that is a uniform contraction, and what survives is the *depth-asymmetric* part of Eq. (4) together with the non-unital (T_1) bias, neither of which a scalar F captures. The cost is a factor two in circuits and the usual variance inflation of any rescaling, so the mitigated curve is not uniformly lower: it trades bias for noise, and helps exactly where bias dominates.

8.2 Shallow versus deep

The practical claim for the adaptive loop is that its shallow circuits should tolerate more noise. We test it directly, and the answer is only half yes.

A first-order Trotterised loop of the same task, with the same ground-state preparation and the same ancilla interferometer, uses $\delta t = 0.05$ — the largest step meeting the method’s own 10^{-3} accuracy target at the reference configuration ($U = 0$ non-trivial), applied unchanged to all four — and needs 62400 controlled CNOTs at $U = 0$ and 68800 at $U = 10$, against 210 and 1674 for the variational readout: a gate-count ratio of 297 and 41. It is also structurally simpler, because it implements the physical propagator including its global phase and therefore needs no φ_G bookkeeping at all: it is exposed to only one of our three legs, the one the two-quadrature estimator already protects.

At $U = 0$ the depth argument wins as advertised: the variational threshold 0.303 beats the Trotter threshold 0.01 by $30\times$ in the non-trivial regime and $13\times$ in the trivial one (Fig. 7(b), Table 2). At $U = 10$ it does not, and it fails in both regimes alike: the Trotterised loop reaches 0.00672 and 0.00729 against 0.00382 and 0.00429 for the variational one, i.e. the $40\times$ deeper circuit is $1.8\times$ and $1.7\times$ more noise-tolerant. That the inversion reproduces in both topological regimes, with thresholds agreeing to 10% within each method, rules out a coincidence of one configuration. The reason is exactly the mechanism of Sec. 6. Trotter pays for its depth only through contrast, and contrast is what the estimator is immune to; the variational route pays through an amplified $\varphi_{G,1}$, and nothing is immune to that. Two caveats keep this honest: the Trotter noiseless deviations (9.3×10^{-4} , 6.5×10^{-3} , 9.8×10^{-3} and 3.1×10^{-3} across the four configurations) are larger than the variational ones (7.7×10^{-5} , 2.6×10^{-5} , 1.1×10^{-3} and 1.6×10^{-4}) — the largest, 9.8×10^{-3} , exceeds that target, since the step was fixed at one configuration and reused — though all still sit a factor 32–337 inside the tolerance, so the baseline is being given the benefit of the coarser step (see `attempts_log.md`, §C2); and the comparison is at one loop time and one system size. The conclusion we draw is not that Trotterisation is preferable — its $\sim 7 \times 10^4$ -gate circuits are far outside what any near-term device will run — but that the shallow-circuit argument for the adaptive method is only as good as the bookkeeping attached to it, and that closing the $\varphi_{G,1}$ gap is worth more than further compressing the ansatz.

8.3 A second device profile

Repeating the sweep against trapped-ion-class rates (TI-B) moves the threshold in η but not the ordering (Fig. 7(c)). The physical reason is visible in the per-channel decomposition: on TI-B the relaxation times are seven orders of magnitude longer than the gate times, so T_1/T_2 drops out entirely and the threshold is set by two-qubit depolarising alone. Quoted in device-referenced units the two profiles agree far better than they do in η : the threshold rises by $2.70\times$ in η for the interacting non-trivial configuration, but only by $2.70 \times 0.75 = 2.0\times$ in ϵ_{2q} once the different base two-qubit rates are accounted for, and that residual factor is precisely the relaxation contribution that TI-B

removes. The practical statement is that what matters is the two-qubit error per CNOT, the number of CNOTs, and the ratio of the circuit duration to T_2 — not the platform label.

The absolute numbers are worth stating plainly, because on this profile one of them crosses into present-day territory. The non-interacting non-trivial configuration reaches $\eta^* = 1.28$ on TI-B — a threshold *above* the base device, i.e. the 136-CNOT loop would return the correct quantised Berry phase on an unmodified trapped-ion processor of that class, with margin. The interacting configuration still needs $\eta^* = 0.0103$, two orders of magnitude better than any device shipping today. The gap between those two statements is the whole content of the depth dependence.

9 Computational cost

Circuits per Berry phase. The measurement cost of the readout is not the interferometer; it is φ_G . With N_θ variational parameters, n_g commuting Hamiltonian groups and M nodes per leg, one Berry-phase estimate requires

$$N_{\text{circ}} = \underbrace{2}_{\text{two quadratures}} + 2(M+1)(N_\theta + n_g) \quad (5)$$

distinct circuit executions, each with S shots. At $M = 48$, $n_g = 10$ (11 at $U = 10$) and $S = 10^4$ this is 3.0×10^3 circuits and 3.0×10^7 shots for the shallow configurations, and 1.8×10^4 circuits and 1.8×10^8 shots for the deep ones — one to a few hours of device time at 10^4 shots per second, dominated entirely by the N_θ connection-overlap circuits. This is the practical bottleneck the depth argument does not capture: the adaptive loop buys two to three orders of magnitude in gates per circuit and pays part of it back in circuits per estimate, because φ_G has one integrand component per variational parameter. Shot-frugal allocation [28, 29] applied preferentially to the dynamical leg is the obvious lever, and Sec. 8 shows that leg is also where mitigation pays.

Classical simulation cost. All results come from full 2^8 - and 2^9 -dimensional density-matrix simulations on CPU. The noiseless anchor sweep (40 trajectories) took ~ 25 CPU-minutes; the noise study is 214 sweep points, each running $2(M+1)$ noisy circuits plus, where mitigation is enabled, the same number again for the noise-estimation circuits, for a total of ~ 20 CPU-hours. Per point, the deterministic circuit simulation is done once and all 32 seeds and all eight shot counts are drawn from the resulting exact output distributions, which is exact and makes the seed and shot axes essentially free.

10 Related work

Adaptive variational dynamics. AVQDS grows a pseudo-Trotter ansatz under a McLachlan-distance threshold [30, 31, 9], with AVQITE supplying the ground state [19]; both descend from ADAPT-VQE [32, 33], and the disjoint-support layer growth used here is the TETRIS construction [20]. The closest methodological antecedent is AVQDS(T) [10], which combines that layering with an eigenvalue truncation of the parameter equation of motion explicitly motivated by noise resilience and reports simulations with realistic channels. Our regularisation choice is in the same spirit; the difference in object is complete, since that work studies quench dynamics of spin models and we study a closed-loop geometric invariant whose error budget is dominated by a cancellation that has no analogue there. The same group’s analysis of sampling noise and optimal measurement distributions in variational imaginary-time evolution [29] is the natural companion to our shot-floor analysis, and the finding that adaptive layering reduces noise susceptibility in ADAPT-type algorithms [34] is the static-energy analogue of the depth dependence we measure.

Berry phases on quantum hardware. The cyclic-adiabatic + ancilla-interferometry route we characterise was introduced in gate-based form by Murta *et al.* [7] with Trotterised evolution, and the underlying primitive is the Ekert interferometer [8]. Alternatives include VQE-based non-adiabatic-coupling and Berry-phase estimators [35] and adiabatic Chern-number protocols [36]. Topological invariants have been extracted from noisy processors in several settings [37, 38, 39], usually with the qualitative claim that the invariants are robust; the quantitative version of that claim has been missing for the adaptive variational route.

Convergent and adjacent work. Two recent preprints overlap our territory and we differentiate explicitly. Kiumi [40] combines forward and time-reversed ($\pm\hat{H}$) evolutions to cancel the dynamical phase and uses a randomised interferometric readout over the full $[0, 2\pi)$ range — structurally the same two ingredients we rely on — but the errors that work removes are *algorithmic* (non-adiabatic, $O(T^{-1})$), and it contains no device-noise model, no shots and no adaptive ansatz; the two studies are on orthogonal error axes and are complementary. Chen *et al.* [41] give gate-based circuits for the many-body Berry phase of the same SSH family, Trotterised and noiseless. On the hardware side, Chang *et al.* [42, 43] measure Berry phases of SSH chains on IBM devices with global-depolarising mitigation, and argue from a Lindblad analysis that dissipation does not affect the invariants. Our results sharpen that: the interferometric leg really is insensitive to a common depolarising contrast — which is what a two-quadrature readout of a noninteracting SSH phase probes — but the variational protocol’s second leg is not, and that is where the finite threshold comes from. Independent convergence on the depolarising-rescaling idea [25] and on readout unfolding [26, 27] is acknowledged; we use both as published and claim no novelty for them.

Noise in variational algorithms. Local Pauli noise flattens variational landscapes exponentially in depth [44], the general framing being [45, 46]; measurement cost is a first-class obstacle [47, 48, 49, 50]; and error mitigation [51, 52, 53, 54, 55, 56] buys accuracy at a sampling cost. The device-noise composition we inject (depolarising + relaxation + readout) follows the model validated against hardware calibration data in [23].

11 Limitations and outlook

Scope of the threshold. The variational trajectory is frozen at its noiseless values, so what we bound is the resilience of the *measured* Berry phase $\varphi_B = \varphi_{qc} + \varphi_G$ at a fixed trajectory. Feeding sampling noise back into the McLachlan solve, so that M, V, θ and the growth decisions all become stochastic, is a strictly harder study that we scope but do not run (`attempts_log.md`, §C1); its cost is 10^4 – 10^5 noisy circuit evaluations per RK4 step. A stochastic trajectory can only add error, so our thresholds should be read as upper bounds for the full pipeline.

System size and loop time. One chain length ($N = 4$, eight qubits), one loop time ($T = 20$) and one pair of dimerisations enter the noise study, although the noiseless anchor covers the whole dimerisation axis. The mechanism we identify is structural rather than size-specific — the amplification factor is the ratio of the half-loop dynamical phase to its residue, $\sim ET/2$ over $O(1)$, and E is extensive — so we expect the dynamical leg to become *more* dominant, not less, at larger N and larger T . Testing that is the obvious next step.

Noise-model composition. Four incoherent families co-vary along a single scalar in the ratio a fixed device profile sets; a device whose ratios differ will have a different threshold in η even at the same ϵ_{2q} , which is why every threshold is also quoted in device units and why the per-channel isolation is reported. Coherent over-rotation and crosstalk are not implemented and not swept (`attempts_log.md`, §C3); a coherent over-rotation is a systematic distortion shared by both halves of the loop, so by the argument of Sec. 6 it should partly cancel in $\varphi_{G,1}$, but we have not measured that. Connectivity is all-to-all, matching the resource model of the method being characterised; a limited-connectivity compilation would add SWAP overhead and lower the thresholds further.

Simulation approximations. The noisy density-matrix evolution is exact; two approximations sit around it. Idle relaxation is applied once per circuit layer with each qubit’s residual idle time rather than after every elementary slice, which is accurate to second order in the per-slice damping ($\leq 10^{-3}$ everywhere here). And the noisy integrands feeding φ_G are simulated at $M = 48$ nodes per leg and interpolated onto the RK4 grid; this is exact at zero noise by construction, since the interpolated quantity is the measured *deviation* from the noiseless integrand and that deviation vanishes identically. Probed at 10^9 shots with all channels off, the residual at $M = 48$ is at most 1.2×10^{-3} rad (`logs/quadrature_check.json`) and does not fall further with M : it is the finite-shot floor of the probe, not a quadrature error, so the node count is not the limiting approximation. The finite M does mean the reported shot noise in φ_G is that of a readout measuring at M nodes per leg, which we state in the protocol and count in the shot budget.

Reproducibility. The implementation (Pauli algebra and Jordan–Wigner mapping, exact diagonalisation and the Wilson-loop reference, the McLachlan solver and adaptive growth, AVQITE, AVQDS, the circuit compiler, the density-matrix noise model, the shot sampler, the Trotter baseline, the mitigation routines and the threshold analysis) is in `code/`, together with the three validation suites. Every sweep point, including the per-seed values behind every figure and table, is in `logs/`; the analysis protocol was written to `logs/protocol.json` before any noisy run. `codebase_notes.md` records what was inspected in the published archive and what was reused; `attempts_log.md` records the disclosed deviations, the constructions that failed, and the studies that were scoped but not run.

References

- [1] M. V. Berry. Quantal phase factors accompanying adiabatic changes. *Proc. R. Soc. Lond. A*, 392:45–57, 1984.
- [2] J. Zak. Berry’s phase for energy bands in solids. *Phys. Rev. Lett.*, 62:2747–2750, 1989.
- [3] Raffaele Resta. Macroscopic polarization in crystalline dielectrics: the geometric phase approach. *Rev. Mod. Phys.*, 66:899–915, 1994.
- [4] Di Xiao, Ming-Che Chang, and Qian Niu. Berry phase effects on electronic properties. *Rev. Mod. Phys.*, 82:1959–2007, 2010.
- [5] Q. Niu, D. J. Thouless, and Y.-S. Wu. Quantized Hall conductance as a topological invariant. *Phys. Rev. B*, 31:3372–3377, 1985.
- [6] Salvatore R. Manmana, Andrew M. Essin, Reinhard M. Noack, and Victor Gurarie. Topological invariants and interacting one-dimensional fermionic systems. *Phys. Rev. B*, 86:205119, 2012.
- [7] Bruno Murta, G. Catarina, and J. Fernández-Rossier. Berry phase estimation in gate-based adiabatic quantum simulation. *Phys. Rev. A*, 101:020302, 2020.
- [8] Artur K. Ekert, Carolina Moura Alves, Daniel K. L. Oi, Michał Horodecki, Paweł Horodecki, and L. C. Kwek. Direct estimations of linear and nonlinear functionals of a quantum state. *Phys. Rev. Lett.*, 88:217901, 2002.
- [9] Yong-Xin Yao, Niladri Gomes, Feng Zhang, Cai-Zhuang Wang, Kai-Ming Ho, Thomas Iadecola, and Peter P. Orth. Adaptive variational quantum dynamics simulations. *PRX Quantum*, 2:030307, 2021.
- [10] Feng Zhang, Cai-Zhuang Wang, Thomas Iadecola, Peter P. Orth, and Yong-Xin Yao. Adaptive variational quantum dynamics simulations with compressed circuits and fewer measurements. *Phys. Rev. B*, 111:094310, 2025.
- [11] Martin Mootz and Yong-Xin Yao. Efficient Berry phase calculation via adaptive variational quantum computing approach. *APL Quantum*, 3(1):016111, 2026.
- [12] Gabriele De Chiara and G. Massimo Palma. Berry phase for a spin 1/2 particle in a classical fluctuating field. *Phys. Rev. Lett.*, 91:090404, 2003.
- [13] A. Carollo, I. Fuentes-Guridi, M. França Santos, and V. Vedral. Geometric phase in open systems. *Phys. Rev. Lett.*, 90:160402, 2003.
- [14] Paolo Zanardi and Mario Rasetti. Holonomic quantum computation. *Phys. Lett. A*, 264:94–99, 1999.
- [15] S. Filipp, J. Klepp, Y. Hasegawa, C. Plonka-Spehr, U. Schmidt, P. Geltenbort, and H. Rauch. Experimental demonstration of the stability of Berry’s phase for a spin-1/2 particle. *Phys. Rev. Lett.*, 102:030404, 2009.
- [16] R. K. L. Colmenar, Utkan Güngördü, and J. P. Kestner. Conditions for equivalent noise sensitivity of geometric and dynamical quantum gates. *PRX Quantum*, 3:030310, 2022.
- [17] W. P. Su, J. R. Schrieffer, and A. J. Heeger. Solitons in polyacetylene. *Phys. Rev. Lett.*, 42:1698–1701, 1979.
- [18] Bing-Tian Ye, Liang-Zhu Mu, and Heng Fan. Entanglement spectrum of Su-Schrieffer-Heeger-Hubbard model. *Phys. Rev. B*, 94:165167, 2016.
- [19] Niladri Gomes, Anirban Mukherjee, Feng Zhang, Thomas Iadecola, Cai-Zhuang Wang, Kai-Ming Ho, Peter P. Orth, and Yong-Xin Yao. Adaptive variational quantum imaginary time evolution approach for ground state preparation. *Adv. Quantum Technol.*, 4:2100114, 2021.

- [20] Panagiotis G. Anastasiou, Yanzhu Chen, Nicholas J. Mayhall, Edwin Barnes, and Sophia E. Economou. TETRIS-ADAPT-VQE: An adaptive algorithm that yields shallower, denser circuit ansätze. *Phys. Rev. Research*, 6:013254, 2024.
- [21] Jesús Cobos, Joana Fraxanet, César Benito, Francesco di Marcantonio, Pedro Rivero, Kornél Kapás, Miklós Antal Werner, Örs Legeza, Alejandro Bermudez, and Enrique Rico. Real-time dynamics in a (2+1)-D gauge theory: The stringy nature on a superconducting quantum simulator, 2025.
- [22] Muhammad AbuGhanem. IBM quantum computers: evolution, performance, and future directions. *J. Supercomput.*, 81:687, 2025.
- [23] Konstantinos Georgopoulos, Clive Emary, and Paolo Zuliani. Modeling and simulating the noisy behavior of near-term quantum computers. *Phys. Rev. A*, 104:062432, 2021.
- [24] S. A. Moses, C. H. Baldwin, M. S. Allman, et al. A race-track trapped-ion quantum processor. *Phys. Rev. X*, 13:041052, 2023.
- [25] Miroslav Urbanek, Benjamin Nachman, Vincent R. Pascuzzi, Andre He, Christian W. Bauer, and Wibe A. de Jong. Mitigating depolarizing noise on quantum computers with noise-estimation circuits. *Phys. Rev. Lett.*, 127:270502, 2021.
- [26] Sergey Bravyi, Sarah Sheldon, Abhinav Kandala, David C. McKay, and Jay M. Gambetta. Mitigating measurement errors in multiqubit experiments. *Phys. Rev. A*, 103:042605, 2021.
- [27] Paul D. Nation, Hwajung Kang, Neereja Sundaresan, and Jay M. Gambetta. Scalable mitigation of measurement errors on quantum computers. *PRX Quantum*, 2:040326, 2021.
- [28] Andrew Arrasmith, Lukasz Cincio, Rolando D. Somma, and Patrick J. Coles. Operator sampling for shot-frugal optimization in variational algorithms, 2020.
- [29] Feng Zhang, Niladri Gomes, Joshua Aftergood, Thomas Iadecola, Yong-Xin Yao, and Peter P. Orth. Sampling noise and optimal measurement distributions in variational quantum imaginary time evolution, 2026.
- [30] A. D. McLachlan. A variational solution of the time-dependent Schrodinger equation. *Mol. Phys.*, 8(1):39–44, 1964.
- [31] Xiao Yuan, Suguru Endo, Qi Zhao, Ying Li, and Simon C. Benjamin. Theory of variational quantum simulation. *Quantum*, 3:191, 2019.
- [32] Harper R. Grimsley, Sophia E. Economou, Edwin Barnes, and Nicholas J. Mayhall. An adaptive variational algorithm for exact molecular simulations on a quantum computer. *Nat. Commun.*, 10:3007, 2019.
- [33] Ho Lun Tang, V. O. Shkolnikov, George S. Barron, Harper R. Grimsley, Nicholas J. Mayhall, Edwin Barnes, and Sophia E. Economou. Qubit-ADAPT-VQE: An adaptive algorithm for constructing hardware-efficient ansätze on a quantum processor. *PRX Quantum*, 2:020310, 2021.
- [34] Christopher K. Long, Kieran Dalton, Crispin H. W. Barnes, David R. M. Arvidsson-Shukur, and Normann Mertig. Layering and subpool exploration for adaptive variational quantum eigensolvers: Reducing circuit depth, runtime, and susceptibility to noise. *Phys. Rev. A*, 109:042413, 2024.
- [35] Shiro Tamiya, Sho Koh, and Yuya O. Nakagawa. Calculating nonadiabatic couplings and Berry’s phase by variational quantum eigensolvers. *Phys. Rev. Research*, 3:023244, 2021.
- [36] Marcel Niedermeier, Marc Nairn, Christian Flindt, and Jose L. Lado. Quantum computing topological invariants of two-dimensional quantum matter. *Phys. Rev. Research*, 6:043288, 2024.
- [37] Daniel Azses, Rafael Haenel, Yehuda Naveh, Robert Raussendorf, Eran Sela, and Emanuele G. Dalla Torre. Identification of symmetry-protected topological states on noisy quantum computers. *Phys. Rev. Lett.*, 125:120502, 2020.
- [38] Adam Smith, Bernhard Jobst, Andrew G. Green, and Frank Pollmann. Crossing a topological phase transition with a quantum computer. *Phys. Rev. Research*, 4:L022020, 2022.
- [39] Xiao Xiao, J. K. Freericks, and A. F. Kemper. Robust measurement of wave function topology on NISQ quantum computers. *Quantum*, 7:987, 2023.
- [40] Chusei Kiumi. Adiabatic error cancellation in Berry phase estimation, 2026.

- [41] Yu-Cheng Chen, Chun-Yu Chen, Dah-Wei Chiou, and Jheng-Wei Jiang. Adiabatic quantum simulation of the topological Su-Schrieffer-Heeger-Hubbard model, 2026.
- [42] Yu-Cheng Chang, Hsiu-Chuan Hsu, and Chun-Chi Lin. Simulating quench dynamics and topological order of the Su-Schrieffer-Heeger chain on a noisy quantum processor. *Phys. Rev. Research*, 7:013043, 2025.
- [43] Yu-Cheng Chang and Hsiu-Chuan Hsu. Simulating dynamical topological invariants and the Berry phase on IBM quantum computers. *Quantum Inf. Process.*, 21:41, 2022.
- [44] Samson Wang, Enrico Fontana, M. Cerezo, Kunal Sharma, Akira Sone, Lukasz Cincio, and Patrick J. Coles. Noise-induced barren plateaus in variational quantum algorithms. *Nat. Commun.*, 12:6961, 2021.
- [45] M. Cerezo, Andrew Arrasmith, Ryan Babbush, Simon C. Benjamin, Suguru Endo, Keisuke Fujii, Jarrod R. McClean, Kosuke Mitarai, Xiao Yuan, Lukasz Cincio, and Patrick J. Coles. Variational quantum algorithms. *Nat. Rev. Phys.*, 3:625–644, 2021.
- [46] John Preskill. Quantum computing in the NISQ era and beyond. *Quantum*, 2:79, 2018.
- [47] Dave Wecker, Matthew B. Hastings, and Matthias Troyer. Progress towards practical quantum variational algorithms. *Phys. Rev. A*, 92:042303, 2015.
- [48] Jérôme F. Gonthier, Maxwell D. Radin, Corneliu Buda, Eric J. Daskocil, Clena M. Abuan, and Jhonathan Romero. Measurements as a roadblock to near-term practical quantum advantage in chemistry: Resource analysis. *Phys. Rev. Research*, 4:033154, 2022.
- [49] Barnaby van Straaten and Bálint Koczor. Measurement cost of metric-aware variational quantum algorithms. *PRX Quantum*, 2:030324, 2021.
- [50] Johannes Jakob Meyer. Fisher information in noisy intermediate-scale quantum applications. *Quantum*, 5:539, 2021.
- [51] Kristan Temme, Sergey Bravyi, and Jay M. Gambetta. Error mitigation for short-depth quantum circuits. *Phys. Rev. Lett.*, 119:180509, 2017.
- [52] Ying Li and Simon C. Benjamin. Efficient variational quantum simulator incorporating active error minimization. *Phys. Rev. X*, 7:021050, 2017.
- [53] Suguru Endo, Simon C. Benjamin, and Ying Li. Practical quantum error mitigation for near-future applications. *Phys. Rev. X*, 8:031027, 2018.
- [54] Ewout van den Berg, Zlatko K. Mineev, Abhinav Kandala, and Kristan Temme. Probabilistic error cancellation with sparse Pauli–Lindblad models on noisy quantum processors. *Nat. Phys.*, 19:1116–1121, 2023.
- [55] Zhenyu Cai, Ryan Babbush, Simon C. Benjamin, Suguru Endo, William J. Huggins, Ying Li, Jarrod R. McClean, and Thomas E. O’Brien. Quantum error mitigation. *Rev. Mod. Phys.*, 95:045005, 2023.
- [56] Youngseok Kim, Andrew Eddins, Sajant Anand, Ken Xuan Wei, Ewout van den Berg, Sami Rosenblatt, Hasan Nayfeh, Yantao Wu, Michael Zaletel, Kristan Temme, and Abhinav Kandala. Evidence for the utility of quantum computing before fault tolerance. *Nature*, 618:500–505, 2023.